Applied Economics

Difference-in-Differences Estimator

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Winter 2025/26

Introduction

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Four **quasi-experimental methods** to *mimic random treatment assignment:*

1. instrumental variables (IV)
2. Panel Data Methods
3. differences-in-differences (DiD)
4. regression discontinuity designs (RDD)

Panel Data Methods to control for unobserved time-invariant endogeneity c_i.

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + c_i + \epsilon_{it}, \quad i = 1, ..., n \quad t = 1, ..., T.$$

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Differences in Differences estimation requires a specific type of panel data:

- Time variation: all units are observed in two time periods.
- Treatment variation:
  - Period 1: no unit receives treatment
  - Period 2: some units are treated, others remain untreated (control group).

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Example: Causal effect of Minimum Wage on Employment

Card & Krueger (1994)

Does an increase in the minimum wage cause a reduction in employment?

Card and Krueger (1994) estimates the causal effect of a minimum wage (MW) increase on the employment rate in the fast-food industry.

- New Jersey (NJ) increased the state MW in 1992 from \$4.25 to \$5.05 per hour (treatment group)
- Neighboring Pennsylvania (PA) did not change its minimum wage level (control group)
- The authors surveyed 410 fast-food establishments in NJ and eastern PA:
  - First survey before the NJ MW increase (February 1992),
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df <- read.csv("./AE_week10_DD/df_min_wage.csv")
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Motivating Example

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A Panel Data model to control for time-invariant endogeneity:

$$Y_{it} = \beta_0 + \beta_1 X_{1,it} + c_i + \epsilon_{it}, \qquad i = 1, ..., n \qquad t = 1, 2.$$

Y_{it} Employment level in restaurant i at time t

X_{it} Indicator variable if restaurant i experienced minimum wage increase at time t

c_i Unobserved time-invariant characteristics of restaurant i

ϵ_{it} Unobserved time-varying endogeneity.

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panel_0 <- feols(emptot ~ is_min_wage_up | sheet, data = df)
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$\rightarrow$ Is exogeneity assumption $E[\epsilon_{it}|X_{it}, c_i] = 0$ reasonable? Why?

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Parallel Trends Assumption

- Panel methods control for time-invariant endogeneity c_i, but...
- We need to control for time-variying endogeneity ϵ_{it}.

How would employment have evolved over time in NJ, *had the minimum wage not been increased*?

Paralell Trends Assumption

$\rightarrow$ Absent treatment, treated units would have followed the same trend as controls.

- In our example: absent minimum wage increase, fast-food employment in NJ would have followed the same trend as in PA.
- Is this assumption reasonable?
  - PA and NJ are very similar: similar institutions, similar habits, similar consumers, similar incomes, similar weather, etc.
- Can we test this assumption with our dataset?

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Visualizing Differences in Differences

DiD Estimator: Basic Form

Start by grouping data into four groups:

```
df_plot <- df %>%
            group_by(state, observation) %>%
            summarise(mean_emptot = mean(emptot, na.rm = TRUE),
                sd_emptot = sd(emptot, na.rm = TRUE),
                .groups = "drop")
```

	NJ
Before Treatment	20.4
After Treatment	21.0
Difference (A-B)	

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	NJ
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After Treatment	21.0
Difference (A-B)	0.6

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	NJ	PA
Before Treatment	20.4	23.3
After Treatment	21.0	21.2
Difference (A-B)	0.6	-2.1

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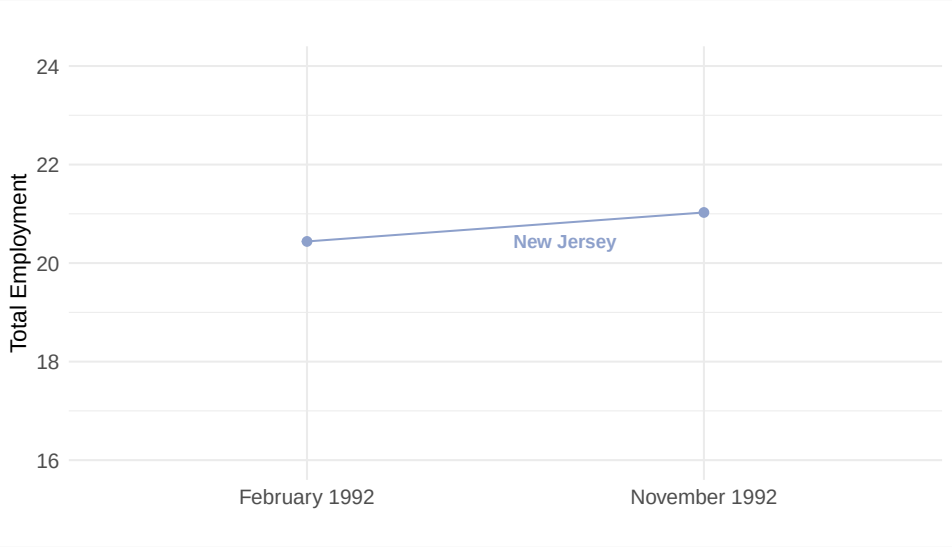
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	NJ	PA	
Before Treatment	20.4	23.3	
After Treatment	21.0	21.2	
Difference (A-B)	0.6	-2.1	DiD estimate: $0.6 - (-2.1) = 2.7$

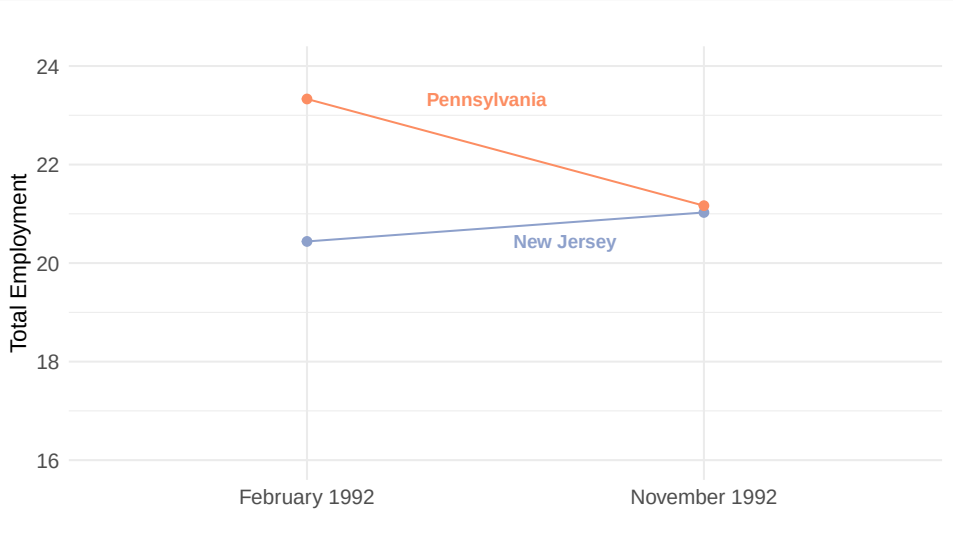
Ussing ggplot2 to Visualize DiD

```
ggplot(df_plot, aes(x = observation, y = mean_emptot,
                    group = state, color = state)) +
geom_line(linewidth = 0.5) +
geom_point(size = 2) +
scale_y_continuous(limits = c(16, 24)) +
theme_minimal()
```

DiD Estimator: Graphical



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Adding the Paralell Trends Assumption

Obtain the time trend (i.e. slope) of Pennsylvania as pa_slope

```
pa_data <- df_plot %>% filter(state == "Pennsylvania")
pa_slope <- diff(pa_data$mean_emptot)
```

Create a new NJ_counterfactual dataset:

```
NJ_counterfactual <- data.frame(
    observation = c("February 1992", "November 1992"),
    mean_emptot = c(nj_feb$mean_emptot, nj_feb$mean_emptot + pa_slope),
    state = "Counterfactual")
```

Plot the counterfactual trend as an **additional line** of the previous code:

```
geom_line(data = NJ_counterfactual,
          aes(x = observation, y = mean_emptot, group = state),
          linetype = "dashed", linewidth = 0.5, color = "gray40") + ...
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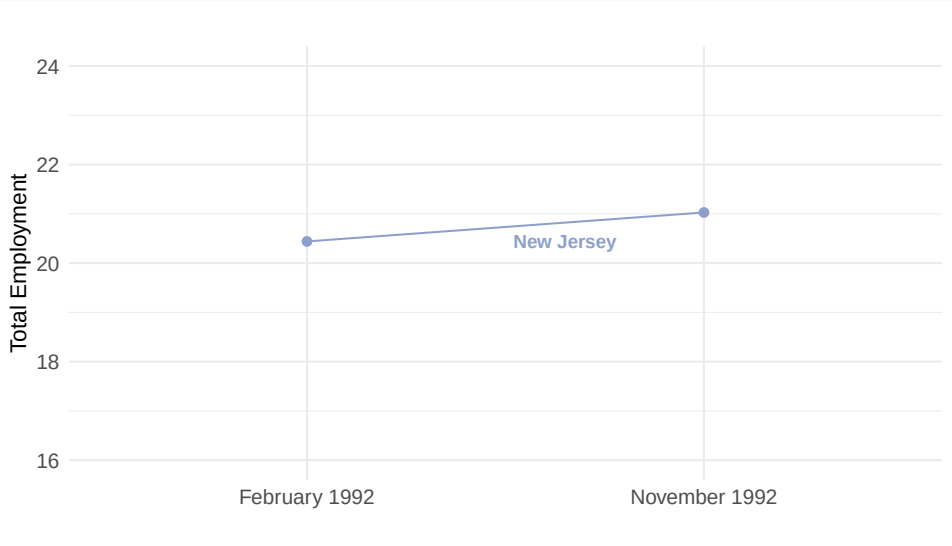
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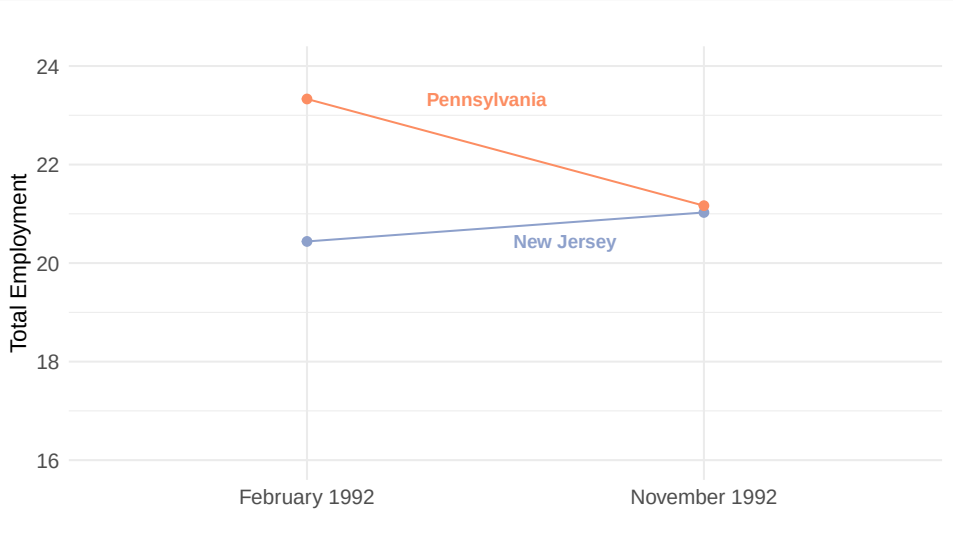
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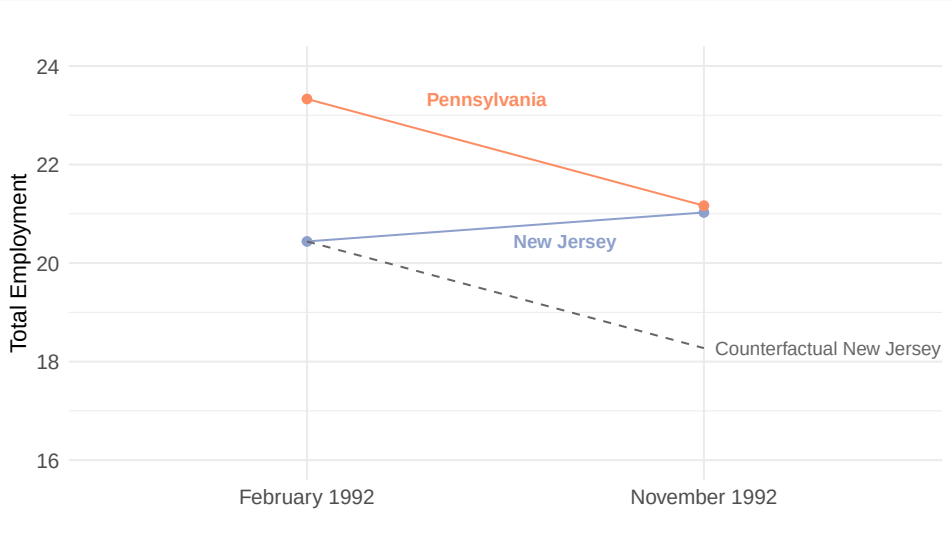
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Diff-in-Diff Estimator: Basic Form

DiD Estimator: Basic Form

More generally, the DiD estimator is given as:

$$\begin{aligned}\widehat{\beta}^{\text{DiD}} &= (\overline{Y}^{\text{treatment,after}} - \overline{Y}^{\text{treatment,before}}) - (\overline{Y}^{\text{control,after}} - \overline{Y}^{\text{control,before}}) \\ &= \Delta\overline{Y}^{\text{treatment}} - \Delta\overline{Y}^{\text{control}}\end{aligned}$$

with

$\overline{Y}^{\text{treatment,before}}$ sample average in the TG before the treatment $\rightarrow \overline{Employm}^{\text{NJ,before}}$

$\overline{Y}^{\text{treatment,after}}$ sample average in the TG after the treatment $\rightarrow \overline{Employm}^{\text{NJ,after}}$

$\overline{Y}^{\text{control,before}}$ sample average in the CG before the treatment $\rightarrow \overline{Employm}^{\text{PA,before}}$

$\overline{Y}^{\text{control,after}}$ sample average in the CG after the treatment $\rightarrow \overline{Employm}^{\text{PA,after}}$

Parallel Trends Assumption

The crucial assumption to identify the causal effect of interest with the DiD estimator is the **Common or Parallel Trends Assumption:**
In the absence of treatment, time trends in the treatment group would have been the same as in the control group.

Example: absent any minimum wage increase, Pennsylvania's fast-food employment trend would have been what we should have expected to see in New Jersey. $\rightarrow$ trend in PA is a reliable counterfactual for NJ

It is impossible to validate whether the assumption holds. (By definition, we cannot observe what would have happened in the absence of treatment in the TG)
Intuitive check: compare trends *before* policy change

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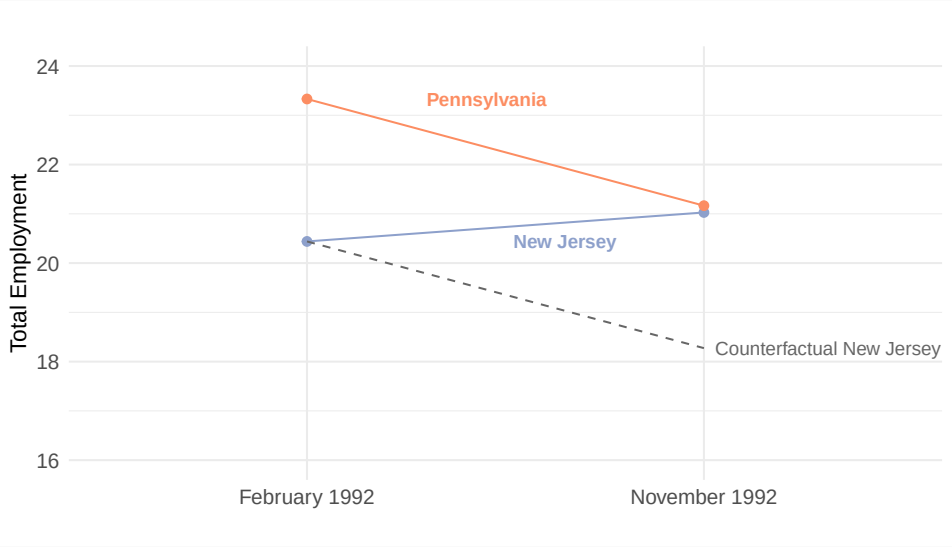
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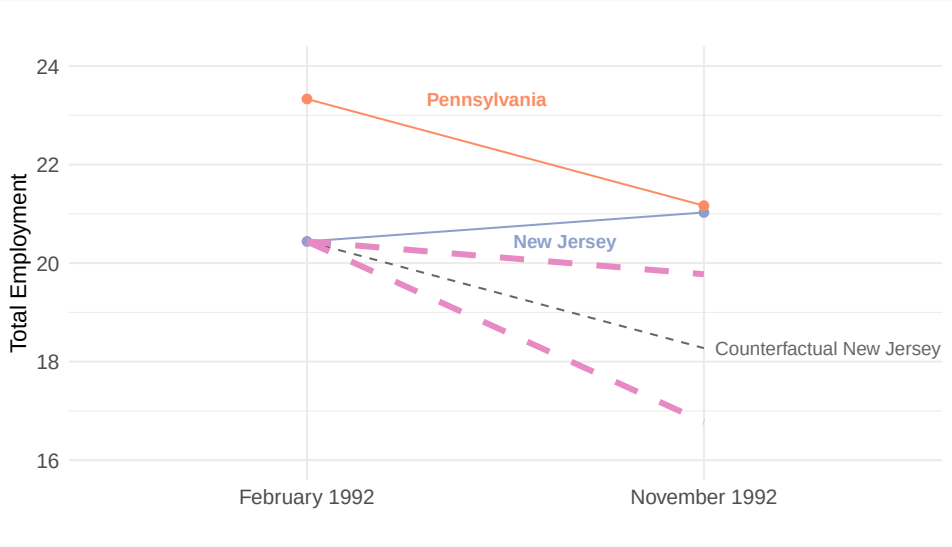
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Visualizing the Parallel Trends Assumption



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Diff-in-Diff Estimator: Regression Form

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The simplets version of the DiD regression model is:

$$Y_{it} = \alpha + \gamma Treat_i + \delta Post_t + \beta_{TE}(Post_t \times Treat_i) + \varepsilon_i$$

```
did_model <- lm(emptot ~ time + treated + time:treated, data = df)
```

$Treat_i$ Treatment group dummy, with index i for cross-sectional units

$$Treat_i = \begin{cases} 0 & \text{if } i \text{ in Pennsylvania,} \\ 1 & \text{if } i \text{ in New Jersey} \end{cases}$$

$Post_t$ Treatment periods dummy, with index t for time periods

$$Post_t = \begin{cases} 0 & \text{if } t < \text{ April 1992,} \\ 1 & \text{if } t \geq \text{ April 1992} \end{cases}$$

$Treat_i \times Post_t$ Interaction term $\rightarrow$ the coefficient β_{TE} is the DiD causal effect!

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Understanding the Regression

$$Y_{it} = \alpha + \gamma Treat_i + \delta Post_t + \beta_{TE}(Post_t \times Treat_i) + \varepsilon_i$$

Expeted outcomes in control units:

- $E(Y_{it}|Treat_i = 0, Post_t = 0) = \alpha$
- $E(Y_{it}|Treat_i = 0, Post_t = 1) = \alpha + \delta$

Expected outcomes in treated units:

- $E(Y_{it}|Treat_i = 1, Post_t = 0) = \alpha + \gamma$
- $E(Y_{it}|Treat_i = 1, Post_t = 1) = \alpha + \gamma + \delta + \beta_{TE}$

The causal effect of the treatment on the treated, without the time trended is β_{TE}

$$\underbrace{(\alpha + \gamma + \delta + \beta_{TE}) - (\alpha + \gamma)}_{\text{Effect of treatment on treated}} - \underbrace{(\alpha - \alpha + \delta)}_{\text{Time trend}} = \beta_{TE}$$

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Diff-in-Diff Estimator: Additional Controls

- Add controls X_i that measure pre-treatment characteristics of the units:

$$Y_{it} = \alpha + \gamma Treat_i + \delta Post_t + \beta_{TE}(Post_t \times Treat_i) + \pi X_i + \varepsilon_i$$

The additional regressors may lead to a more precise estimate of β_{TE}.

- Add unit-level fixed effects FE_i to control for all time-invariant unobservables:

$$Y_{it} = \alpha + \gamma Treat_i + \delta Post_t + \beta_{TE}(Post_t \times Treat_i) + \sum_i \pi_i FE_i + \varepsilon_i$$

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panel_1 <- feols(emptot ~ time + treated + is_min_wage_up |
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summary(panel_1, vcov = "hetero")
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Summary

- DiD is a widely used quasi-experimental method to estimate causal effects when treatment is not randomly assigned.
- It requires panel data on a treatment and a control group, as well as data for at least two time periods (before and after treatment).
- The key identifying assumption is the Parallel Trends Assumption: in the absence of treatment, time trends in the treatment group would have been the same as in the control group.
- The DiD estimator can be implemented in a regression framework, which allows to easily extend the basic model.